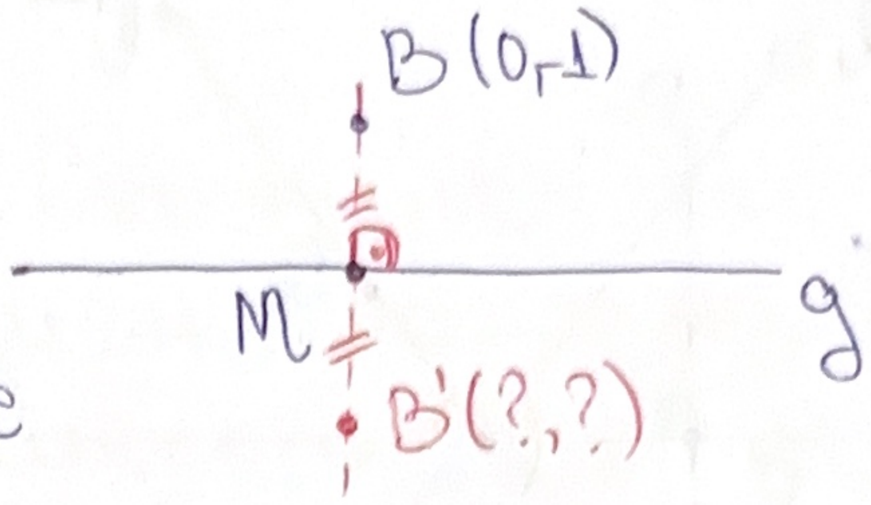
4зад. (Симетрия относно права)

Дадени са права $g: x+y-1=0$ и т. $B(0,-1)$

σ_g - симетрия относно g

т. $B \xrightarrow{\sigma_g}$ т. $B' \Leftrightarrow g \equiv s_{BB'}$ $(\sigma_g(B)=B')$



Да се намерят координатите на точка B'.

реш.: 1) Нека $h \begin{cases} \ni B(0,-1) \\ \perp g: x+y-1=0 \Rightarrow \vec{n}_g(1,1) \end{cases}$

$\Rightarrow h \begin{cases} \ni B(0,-1) \\ \parallel \vec{n}_g(1,1) \end{cases} \Rightarrow h: x-y+C=0 \Big\} \Rightarrow 0-(-1)+C=0 \Rightarrow C=-1$

$\Rightarrow h: x-y-1=0$

2) Нека $h \cap g = M$

$\left| \begin{array}{l} x-y-1=0 \\ x+y-1=0 \end{array} \right. \Rightarrow \left| \begin{array}{l} 2x=2 \\ x-y-1=0 \end{array} \right. \Rightarrow \left| \begin{array}{l} x=1 \\ y=0 \end{array} \right. \Rightarrow M(1,0)$

3) $B(0,-1)$, Нека $B'(x',y')$

$M(1,0)$ - среда на $BB' \Rightarrow \left| \begin{array}{l} \frac{x'+x_B}{2}=1 \\ \frac{y'+y_B}{2}=0 \end{array} \right.$

$\Rightarrow \left| \begin{array}{l} x'+0=2 \\ y'+(-1)=0 \end{array} \right. \Rightarrow \left| \begin{array}{l} x'=2 \\ y'=1 \end{array} \right. \Rightarrow B'(2,1)$

$B(0,-1) \xrightarrow{\sigma_g} B'(2,1)$

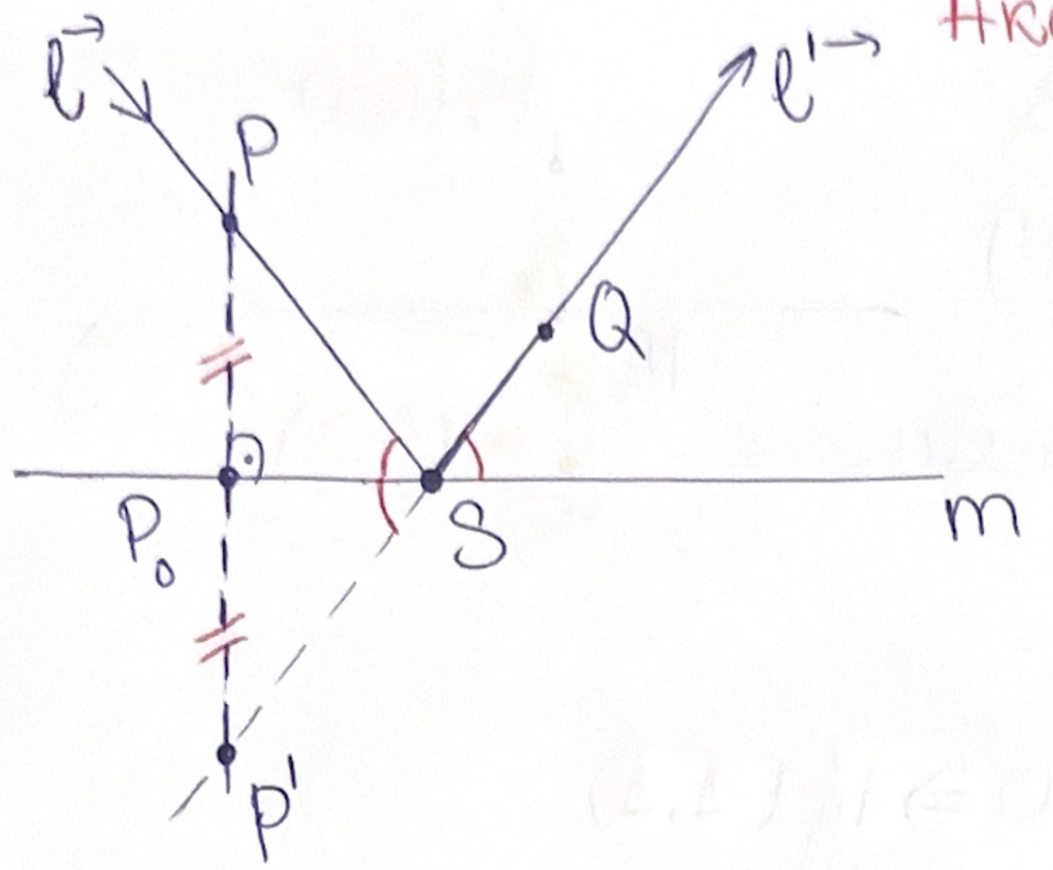
5зад. (Светлинни лъчи)

Дадени са права $m: x+y-3=0$, т. $P(-5,4)$, т. $Q(-1,1)$

Светлинен лъч $\vec{l}$ минава през т. P, отразява се от правата m и отразеният лъч $\vec{l'}$ минава през т. Q. Намерете у-я на правите l и l',

кошто свеждат пъсшше.

Реш.:



Ако т. $P \xrightarrow{a_m} P'$, то $P' \in l'$!

1) Търсим $P'(?,?)$:

Нека $h \perp l$, $P(-5,4)$

l: $x+y-3=0$

$\vec{n}_m (1,1)$

$h \perp l$, $P(-5,4)$

$\parallel \vec{n}_m (1,1) \Rightarrow h: x-y+C=0$

$-5-4+C=0$

$\Rightarrow C=9$

$\Rightarrow h: x-y+9=0$

Нека т. $P_0 = h \cap m$

$$+\begin{cases} x-y+9=0 \\ x+y-3=0 \end{cases} \Rightarrow 2x+6=0 \Rightarrow \begin{cases} x=-3 \\ y=-3+9=6 \end{cases} \Rightarrow P_0(-3,6)$$

P_0 - средата на $PP' \Rightarrow$

$$\begin{cases} -3 = \frac{1}{2}(-5+x') \Rightarrow x' = -1 \\ 6 = \frac{1}{2}(4+y') \Rightarrow y' = 8 \end{cases} \Rightarrow P'(-1,8)$$

$\parallel$ $\parallel$

P_0 P

2) Общо у-е на l':

(1ч) $P'(-1,8)$, $Q(-1,1) \in l' \Rightarrow$

и $M(x,y)$ - произв.

$$\begin{vmatrix} x & y & 1 \\ -1 & 8 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 0 \Rightarrow 8x - y - 1 - (-8) - x - (-y) = 0$$

$$7x + 7 = 0 \Rightarrow l': x+1=0$$

(2ч) $P'(-1,8)$, $Q(-1,1) \Rightarrow l': x=-1$ или $l': x+1=0$

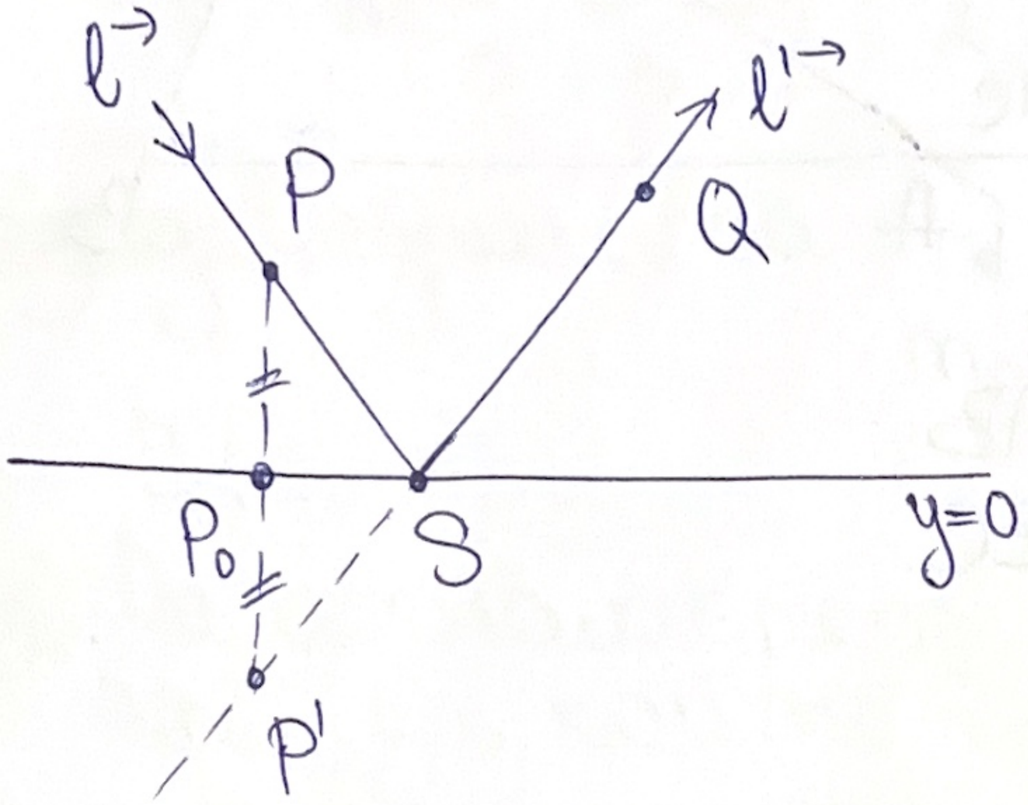
3) $l' \cap m = S \Rightarrow \begin{cases} x+1=0 \\ x+y-3=0 \end{cases} \Rightarrow \begin{cases} x=-1 \\ y=4 \end{cases} \Rightarrow$ т. $S(-1,4)$

4) $l \left| \begin{array}{l} \ni S(-1,4) \\ \ni P(-5,4) \end{array} \right. \Rightarrow l: y=4$

Задача (за упр.)

Дадени са т. $P(2,4)$ и т. $Q(0,1)$. Светлинен лъч $l^{\rightarrow} \ni P$, отразява се от абцисната ос $Ox (y=0)$ и отразеният лъч $l'^{\rightarrow} \ni Q$. Намерете у-я на правите l и l'.

Реш.: 1) Нека т. $P \xrightarrow{\sigma_{Ox}}$ т. $P' \Rightarrow P' \in l'$



Търсим $P'(?,?)$:

Нека $h \left| \begin{array}{l} \ni P(2,4) \\ \perp Ox: y=0 \Rightarrow \vec{n_x}(0,1) \end{array} \right.$

$\Rightarrow h: \left| \begin{array}{l} \ni P(2,4) \\ \parallel \vec{n_x}(0,1) \end{array} \right.$

$\Rightarrow h: x+C=0 \Rightarrow C=-2$

$h: x-2=0$

Нека $h \cap Ox = P_0 \Rightarrow \left| \begin{array}{l} x=2 \\ y=0 \end{array} \right. \Rightarrow P_0(2,0)$

P_0 - средата на $PP' \Rightarrow \left| \begin{array}{l} 2=\frac{1}{2}(x'+2) \\ 0=\frac{1}{2}(y'+4) \end{array} \right. \Rightarrow \begin{array}{l} x'=2 \\ y'=-4 \end{array}$

$P'(2,-4)$

2) $l' \left| \begin{array}{l} \ni P'(2,-4) \\ \ni Q(0,1) \end{array} \right. \Rightarrow \begin{vmatrix} x & y & 1 \\ 2 & -4 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0 \Rightarrow -4x+2-x-2y=0$

$-5x-2y+2=0$

$\underline{l': 5x+2y-2=0}$

$\forall M(x,y)$ - произв.

3) $l' \cap Ox = S$

$\left| \begin{array}{l} 5x+2y-2=0 \\ y=0 \end{array} \right. \Rightarrow 5x=2 \Rightarrow x=\frac{2}{5} \Rightarrow S\left(\frac{2}{5},0\right)$

4) $\ell \perp\!\!\!\perp P(2,4)$
$\ell \supset S(\frac{2}{5},0)$ $\Rightarrow \ell: \begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ \frac{2}{5} & 0 & 1 \end{vmatrix} = 0 \Rightarrow 4x + \frac{2}{5}y - \frac{8}{5} - 2y = 0$

$\Rightarrow 4x - \frac{8}{5}y - \frac{8}{5} = 0 \mid .5$

$20x - 8y - 8 = 0 \mid :4$

$5x - 2y - 2 = 0$

(Запис)

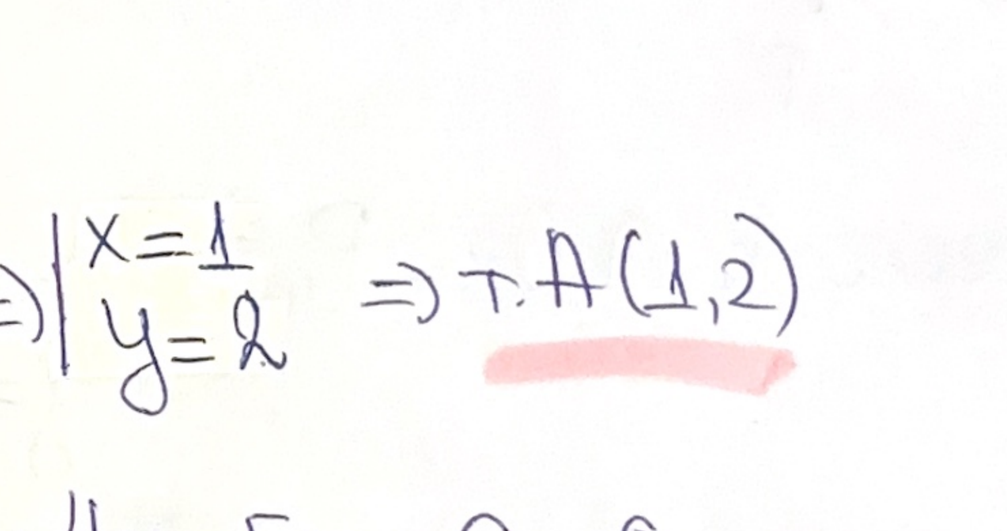
Дадено: $b: 5x + 4y - 13 = 0$

$c: x + 2y - 5 = 0$

$H(14,15)$

Да се намерят координатите на върховете на $\triangle ABC$, ако правите b и c съдържат съответно страните AC и AB, а т. H е ортоцентърът на $\triangle ABC$.

$S_{ABC} = ?$ → за припомняне



Реш.: 1) $b \cap c = A$

$$+\begin{cases} 5x + 4y - 13 = 0 \\ x + 2y - 5 = 0 \mid .(-2) \end{cases} \begin{cases} x = 5 - 2y \\ 3x - 3 = 8 \end{cases} \Rightarrow \begin{cases} x = 1 \\ y = 2 \end{cases} \Rightarrow T.A(1,2)$$

2) $h_b \begin{cases} \perp b: 5x + 4y - 13 = 0 \\ \supset H(14,15) \end{cases} \Rightarrow h_b: 4x - 5y + C = 0$

$\Rightarrow 4.14 - 5.15 + C = 0$

$C = 75 - 56 = 19$

$\Rightarrow h_b: 4x - 5y + 19 = 0$

$h_b \cap c = B \Rightarrow \begin{vmatrix} 4x - 5y + 19 = 0 \\ x + 2y - 5 = 0 \mid .(-4) \end{vmatrix} \Rightarrow \begin{vmatrix} 4x - 5y + 19 = 0 \\ -4x - 8y + 20 = 0 \end{vmatrix}$

$\Rightarrow -13y + 39 = 0 \Rightarrow y = 3 \Rightarrow x = -1 \Rightarrow T.B(-1,3)$

3) $h_c \begin{cases} \perp c: x + 2y - 5 = 0 \Rightarrow h_c: 2x - y + C = 0 \\ \supset H(14,15) \end{cases}$

$2.14 - 15 + C = 0 \Rightarrow C = -13$

$\Rightarrow h_C: 2x-y-13=0$

$h_C \cap b = \{C\}$

$$\begin{vmatrix} 2x-y-13=0 \,/\cdot 4 \\ 5x+4y-13=0 \end{vmatrix} \Rightarrow \begin{vmatrix} 8x-4y-4\cdot 13=0 \\ 5x+4y-13=0 \end{vmatrix}$$

$$13x-5\cdot 13=0$$

$$x=5 \Rightarrow y=2\cdot 5-13=-3$$

$$\Rightarrow T.C(5,-3)$$

$$S_{ABC}=\frac{1}{2}\left\|\begin{matrix} 1 & 2 & 1 \\ -1 & 3 & 1 \\ 5 & -3 & 1 \end{matrix}\right\| = \frac{1}{2}|3+10+3-15-(-2)-(-3)| = \frac{1}{2}\cdot 6 = 3 \text{ кв.ед.}$$

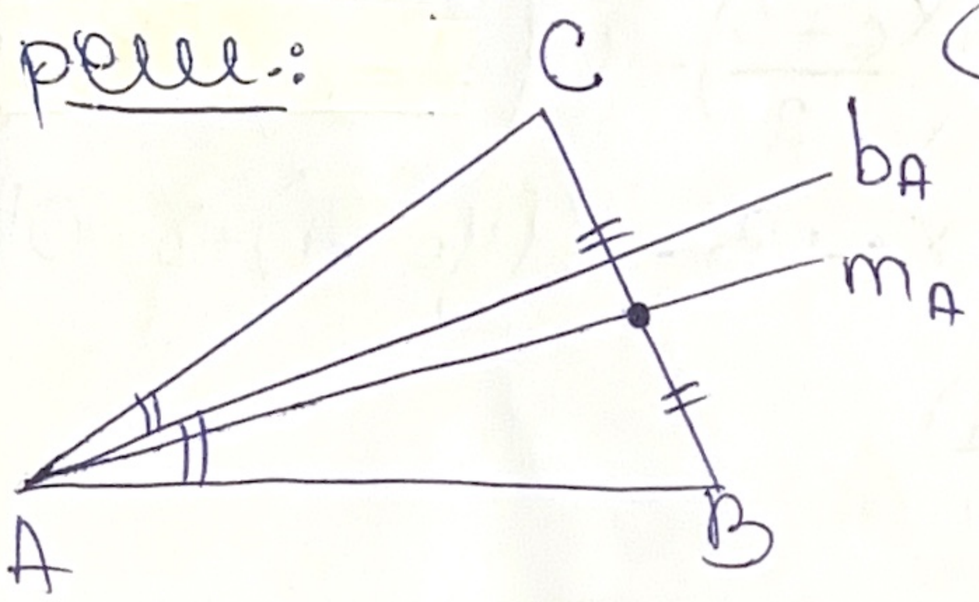
Зад. Дадени са: $b_A: 2x-3y-5=0$

$m_A: x-8y+4=0$

и $B(3,-4)$

а) Да се намерят координатите на A и C на ΔABC, за който b_A и m_A са съотв. вътрешна та ъглополовяща и медианата през върха A

Реш.:



1) $b_A \cap m_A = \{A\}$

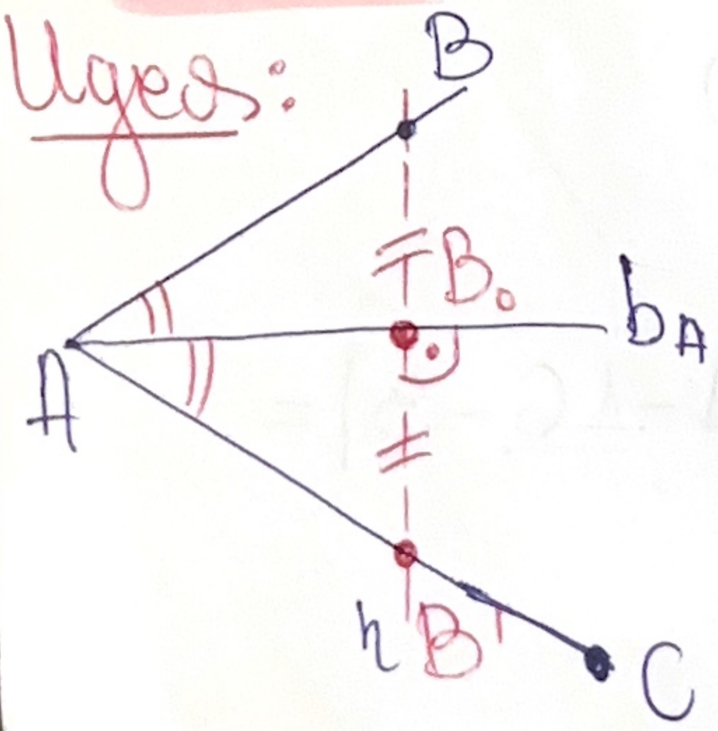
$$\begin{vmatrix} 2x-3y-5=0 \\ x-8y+4=0 \,/\cdot(-2) \end{vmatrix}$$

$$\begin{vmatrix} 2x-3y-5=0 \\ -2x+16y-8=0 \end{vmatrix} \Rightarrow 13y-13=0$$

$$y=1$$

$$\Rightarrow x=4$$

$$\Rightarrow A(4,1)$$



2) Нека $B \xrightarrow{\sigma_{b_A}} B'$. Тогава $B' \in AC$

Нека $h \begin{cases} \ni B(3,-4) \\ \perp b_A: 2x-3y-5=0 \end{cases}$

$\Rightarrow h: 3x+2y+C=0$

$3\cdot 3+2\cdot(-4)+C=0 \Rightarrow C=-1$

$h: 3x+2y-1=0$

$h \cap b_A = B_0$ - среда на BB'

$$\begin{vmatrix} 2x-3y-5=0 & /\cdot(+2) \\ 3x+2y-1=0 & /\cdot 3 \end{vmatrix} \Rightarrow + \begin{vmatrix} 4x-6y-10=0 \\ 9x+6y-3=0 \end{vmatrix}$$

$$13x-13=0 \Rightarrow x=1 \Rightarrow y=-1 \Rightarrow B_0(1,-1)$$

$$\begin{vmatrix} 1=\frac{1}{2}(3+x') \\ -1=\frac{1}{2}(-4+y') \end{vmatrix} \Rightarrow \begin{vmatrix} x'=-1 \\ y'=2 \end{vmatrix} \Rightarrow B'(-1,2)$$

3) $AB': \begin{vmatrix} x & y & 1 \\ -1 & 2 & 1 \\ 4 & 1 & 1 \end{vmatrix} = 0$

$$2x+4y-1-8-x+y=0$$

$$AB': x+5y-9=0$$

4) $\tau. C \in AB' \Rightarrow x_C+5y_C-9=0$ (1)

M - среда на BC

$$\Rightarrow M\left(\frac{x_C+3}{2}, \frac{y_C-4}{2}\right) \in m_A \Rightarrow \left(\frac{x_C+3}{2}\right)-8\cdot\left(\frac{y_C-4}{2}\right)+4=0 \,/\cdot 2$$

$$m_A: x-8y+4=0 \qquad x_C+3-8(y_C-4)+8=0$$

от (1) и (2) $\Rightarrow \begin{vmatrix} x_C+5y_C-9=0 & /\cdot(-1) \\ x_C-8y_C+43=0 \end{vmatrix}$

$$+\begin{vmatrix} -x_C-5y_C+9=0 \\ x_C-8y_C+43=0 \end{vmatrix} \Rightarrow \begin{vmatrix} x_C+5y_C-9=0 \\ -13y_C=-52 \Rightarrow y_C=4 \end{vmatrix} \Rightarrow x_C=-11$$

$$\Rightarrow \tau. C(-11,4)$$

б) $S_{ABC}=?$

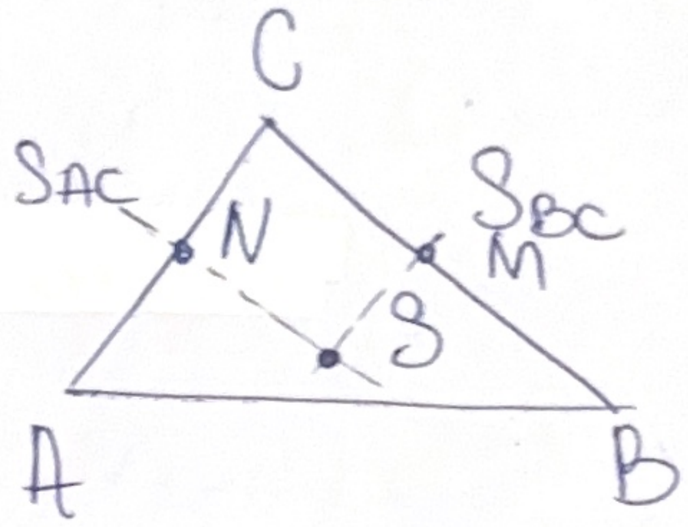
$$S_{ABC}=\frac{1}{2}\left\|\begin{vmatrix} 4 & 1 & 1 \\ 3 & -4 & 1 \\ -11 & 4 & 1 \end{vmatrix}\right\| = \frac{1}{2}|-16-11+12-44-16-3| =$$

$$=\frac{1}{2}|-20-55-3| = \frac{1}{2}\cdot 78 = 39 \text{ кв. ед.}$$

в) Да се намерят координатите на центъра S и дължината на радиуса на описаната окр. около ΔABC.

1н. $S_{AB}=?$, $S_{BC}=?$ (за упр.)

т. $S = S_{AB} \cap S_{BC}$



1) S_{AC} | $\supset N$ – ср. на $AC \Rightarrow N(-\frac{7}{2}, \frac{5}{2})$

| $\perp AC \equiv AB'$: $x+5y-9=0$

$\Rightarrow S_{AC}$: $5x-y+C=0$

$5\cdot(-\frac{7}{2})+(-\frac{5}{2})+C=0 \;/\cdot 2$

$-35-5+2C=0 \Rightarrow C=20$

$\Rightarrow S_{AC}$: $5x-y+20=0$

2) S_{BC}: | M – средата на BC, $M(-4,0)$

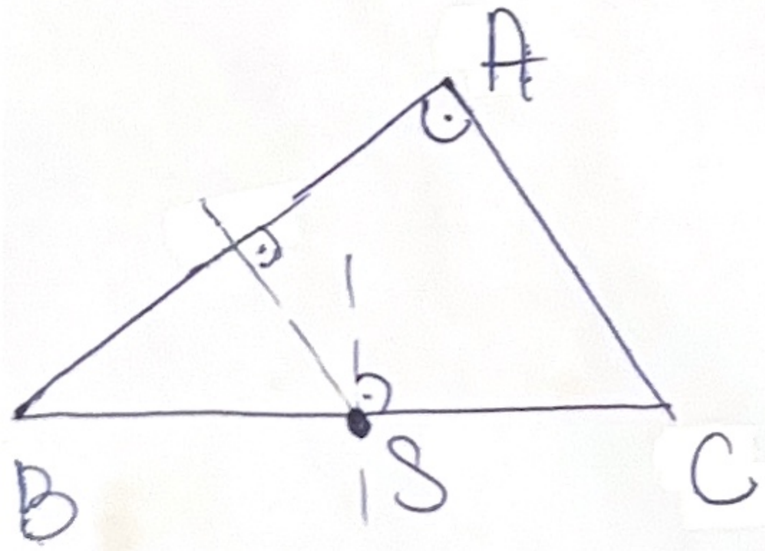
| $\perp BC$: $4x+4y+16=0 \rightarrow$ за упр.

S_{BC}: $4x-4y+28=0$

$S_{AC} \cap S_{BC} =_T S$

$\left|\begin{array}{l} 5x-y+20=0 \;/\cdot(-4) \\ 4x-4y+28=0 \end{array}\right. \Rightarrow \left|\begin{array}{l} -20x+4y-80=0 \\ 4x-4y+28=0 \end{array}\right.$

$-13x-52=0 \Rightarrow x=-4 \Rightarrow y=0 \Rightarrow S(-4,0) \equiv M$



$\Rightarrow R=\frac{1}{2}BC=|\overrightarrow{SB}|=\sqrt{49+16}=\sqrt{65}$

$\overrightarrow{SB}(7,-4)$

(24.) т. $S(x,y)$ – на равни разстояния от A, B и C (сприте вдясно)

$$\left.\begin{array}{l}\overrightarrow{AS}(x-4,y-1)\\ \overrightarrow{BS}(x-3,y+4)\\ \overrightarrow{CS}(x+11,y-4)\end{array}\right\}\Rightarrow |\overrightarrow{AS}|=|\overrightarrow{BS}|=|\overrightarrow{CS}|$$

$$\left|\begin{array}{l}(x-4)^2+(y-1)^2=(x-3)^2+(y+4)^2\\ (x-4)^2+(y-1)^2=(x+11)^2+(y-4)^2\end{array}\right.$$

$$\left|\begin{array}{l}x^2-8x+16+y^2-2y+1=x^2-6x+9+y^2+8y+16\\ x^2-8x+16+y^2-2y+1=x^2+22x+121+y^2-8y+16\end{array}\right.$$

$$\left|\begin{array}{l}-8x-2y+1=-6x+9+8y\\ -8x-2y+1=22x+121-8y\end{array}\right.$$

$$\left|\begin{array}{l}-2x-10y-8=0\ /:(-2)\\ -30x+6y-120=0\ /:(-6)\end{array}\right.\Rightarrow\left|\begin{array}{l}x+5y+4=0\ /\cdot(-5)\\ 5x-y+20=0\end{array}\right.$$

$$+\left|\begin{array}{l}-5x-25y-20=0\\ 5x-y+20=0\end{array}\right.\Rightarrow -26y=0\Rightarrow y=0\Rightarrow x=-4$$

т. $S(-4,0)$

$\overrightarrow{AS}(-8,-1)\Rightarrow |\overrightarrow{AS}|=\sqrt{64+1}=\sqrt{65}$